SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Duration: 60 Mins **QUESTIONS: 5** **Total Marks:50**

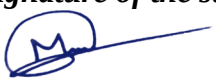
Annexure A

Debugging & Optimisation Task

Code of Conduct:

I MONISH.L [1972260035] _ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Understanding of Concepts	Experimental Design	Use of Tools	Analysis of Results	Documentation	Total
Weightage	25%	30%	20%	15%	10%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty *Total Marks Awarded:*

Annexure A

Debugging & Optimisation Task

1. A network administrator configures a system with IP address **192.168.10.65**, subnet mask **255.255.255.192**, and default gateway **192.168.10.1**, while other systems are configured within the subnet **192.168.10.0/26**. Using the given subnet mask, apply subnetting techniques to determine the network address, broadcast address, and valid host range of the subnet. Verify whether the configured IP address belongs to the given subnet range. Based on your calculations, identify any configuration error related to IP addressing or gateway. Determine the correct IP address and valid gateway that should be assigned to the system. Also, calculate the number of usable host addresses available in the subnet after correction. In your solution, ensure clarity, correctness of calculations, proper step-by-step representation, and accuracy in the final configuration (Code Quality). **(Marks 10)**

2. A company is using the IP block **192.168.1.0/24** and has divided it into subnets using the subnet masks **/26, /27, and /28** for different departments. However, some departments are experiencing IP address shortages while others have unused addresses. Using the given subnet masks, apply subnetting techniques to determine the network address, broadcast address, and valid host range for each subnet. Calculate the number of usable and unused IP addresses in each subnet. Based on your calculations, identify any inefficiencies or mismatches in the current subnet allocation. Suggest corrected or optimized subnet assignments to ensure better utilization of IP addresses within the given network block. **(Marks 10)**

3. A system is configured with the IPv6 address **2001:db8::ff00:42:8329/64**, and another device is configured with **2001:db8:0:0:1::1/64**, but the devices are unable to communicate. Using IPv6 addressing rules, apply expansion to convert both addresses into their full notation. Determine the network prefix of each

address using the given **/64** prefix length. Verify whether both addresses belong to the same network prefix. Based on your calculations, identify any addressing or prefix mismatch causing the issue. Suggest the correct IPv6 configuration required to ensure proper communication and calculate the total number of possible host addresses available within the network. **(Marks 10)**

4. A router is configured with the following routing table entries: **192.168.1.0/24 → next hop** 10.0.0.2, **192.168.2.0/24 → next hop 10.0.0.3**, and a default route **→ 10.0.0.1**, but some packets are not reaching the correct destination. A packet arrives with the destination IP address **192.168.2.100**. Using the given routing table, apply routing techniques to identify the matching network for the destination IP address and determine the next-hop address used to forward the packet. Verify whether the destination IP falls within the subnet **192.168.2.0/24**. Based on your findings, identify any possible routing misconfiguration. Suggest the correct or optimized routing entry to ensure proper packet delivery, and also identify the route that will be used if no matching network entry is found. **(Marks 10)**

5. Two LANs are connected using a router, where LAN1 uses the subnet **192.168.5.0/27** and LAN2 uses **192.168.5.32/27**, but communication between the networks is not functioning properly. A PC in LAN1 is assigned the IP address **192.168.5.10**, and a PC in LAN2 is assigned **192.168.5.40**. Using the given subnet information, apply subnetting techniques to determine the network address, broadcast address, and valid host range for each LAN. Verify whether the given IP addresses belong to their respective subnet ranges. Based on your calculations, identify any possible configuration issues related to addressing or gateway settings. Suggest appropriate gateway IP addresses and optimized configuration changes to ensure successful communication between the two LANs, and calculate the number of usable host addresses available in each subnet. **(Marks 10)**

ANSWERS

Answer 1:

Given subnet mask : 255.255.255.192 = /26 (64 addresses per subnet)

Stated subnet : 192.168.10.0/26

Network address : 192.168.10.0
Broadcast address : 192.168.10.63
Valid host range : 192.168.10.1 – 192.168.10.62

Verification : the configured IP 192.168.10.65 does NOT fall within the range 192.168.10.1 – 192.168.10.62. It actually lies in the next block, 192.168.10.64/26 (range .64 – .127).

Configuration error identified :

• The system's IP address (192.168.10.65) belongs to a different subnet than the one it is meant to be part of (192.168.10.0/26).
• The gateway 192.168.10.1 is valid only for the 192.168.10.0/26 subnet, so a host using .65 cannot reach it — an address/gateway mismatch.

Correction :

Assign the system a valid address from 192.168.10.0/26, e.g. 192.168.10.2 – 192.168.10.62, keeping the gateway as 192.168.10.1.

Usable hosts after correction : 2^6 − 2 = 62 usable host addresses in the /26 subnet.

Answer 2:

Given block : 192.168.1.0/24, subdivided using /26, /27 and /28.

Subnet A → /26 (assigned first)
Network : 192.168.1.0 Broadcast : 192.168.1.63
Valid host range : 192.168.1.1 – 192.168.1.62
Usable hosts : 2^6 − 2 = 62 | Unused in block : 0 (fully sized block)

Subnet B → /27
Network : 192.168.1.64 Broadcast : 192.168.1.95
Valid host range : 192.168.1.65 – 192.168.1.94
Usable hosts : 2^5 − 2 = 30

Subnet C → /28
Network : 192.168.1.96 Broadcast : 192.168.1.111
Valid host range : 192.168.1.97 – 192.168.1.110
Usable hosts : 2^4 − 2 = 14

Remaining unused range : 192.168.1.112 – 192.168.1.255 (144 addresses left unallocated).

Inefficiency identified :

The /28 subnet (14 usable hosts) is too small for a department that is running short of addresses, while the /26 subnet (62 usable hosts) is oversized if its department uses far fewer hosts — causing shortages in one subnet and wastage in another.

Optimized allocation suggested :

Re-size each subnet to match actual host requirements using VLSM (allocate the largest actual requirement first): give the address-starved department a larger block (e.g. /26 or /25 as needed) and shrink over-provisioned departments to /28 or /29. Keep the unused 192.168.1.112 – .255 range as a reserve pool for future growth.

Answer 3:

Device 1 : 2001:db8::ff00:42:8329/64

Expanded form : 2001:0db8:0000:0000:0000:ff00:0042:8329

Device 2 : 2001:db8:0:0:1::1/64

Expanded form : 2001:0db8:0000:0000:0001:0000:0000:0001

Network prefix (/64, first 64 bits = first 4 groups) :

Device 1 prefix : 2001:0db8:0000:0000::/64

Device 2 prefix : 2001:0db8:0000:0000::/64

Verification : both addresses expand to the same network prefix, 2001:db8::/64 — so there is NO prefix mismatch between the two devices; they are correctly placed on the same subnet.

Cause of the communication issue :

Since the /64 prefixes match and the interface IDs (0000:ff00:0042:8329 and 0001:0000:0000:0001) are different (no duplicate address), the fault is not in the IPv6 addressing/subnetting itself. The issue likely lies elsewhere — e.g. a faulty physical/link-layer connection, a firewall or ACL blocking traffic, a missing/incorrect default gateway, or a Neighbor Discovery (ND) failure — and should be investigated at those layers.

Recommended configuration : keep both devices on 2001:db8::/64 with their current unique interface IDs; verify link connectivity, gateway, and firewall rules rather than re-addressing.

Total possible host addresses in the /64 network : 2^(128−64) = 2^64 = 18,446,744,073,709,551,616.

Answer 4:

Routing table : 192.168.1.0/24 → 10.0.0.2 , 192.168.2.0/24 → 10.0.0.3 , default route → 10.0.0.1

Destination IP : 192.168.2.100

Matching network : 192.168.2.0/24 — since .100 lies between .1 and .254 of this subnet.

Next-hop address : 10.0.0.3 (longest-prefix / exact match on 192.168.2.0/24).

Verification : 192.168.2.100 does fall within 192.168.2.0/24, so this entry is the correct match for the packet.

Possible misconfiguration :

If packets are still not reaching the destination even though the table entry is correct, the likely faults are: the next-hop 10.0.0.3 is unreachable/down, a missing or incorrect route to the 10.0.0.0/24 transit network on the router, or a conflicting/overlapping route entry taking precedence over the correct one.

Suggested correction :

Confirm the entry 192.168.2.0/24 → 10.0.0.3 is present and that 10.0.0.3 is reachable (up and responding); remove any duplicate or overlapping routes that might override it.

Default route usage : 0.0.0.0/0 → 10.0.0.1 is used only when a destination does not match either 192.168.1.0/24 or 192.168.2.0/24 in the table.

Answer 5:

LAN1 – 192.168.5.0/27

Network address : 192.168.5.0 Broadcast address : 192.168.5.31

Valid host range : 192.168.5.1 – 192.168.5.30

PC1 (192.168.5.10) verification : within range – valid.

LAN2 – 192.168.5.32/27

Network address : 192.168.5.32 Broadcast address : 192.168.5.63

Valid host range : 192.168.5.33 – 192.168.5.62

PC2 (192.168.5.40) verification : within range – valid.

Issue identified :

Both host addresses are correctly within their own subnets, so the fault is not the host addressing itself. No default gateway has been specified for either PC — without a gateway configured (and matching a router interface in each subnet), the PCs cannot route traffic to the other LAN, which explains the communication failure.

Suggested gateway configuration :

LAN1 gateway : 192.168.5.1 (first usable address, configured on the router interface for LAN1); set as the default gateway on PC1.
LAN2 gateway : 192.168.5.33 (first usable address, configured on the router interface for LAN2); set as the default gateway on PC2.

Usable hosts per subnet : 2^5 − 2 = 30 usable host addresses in each /27 LAN.